Radical Pair System Comparison

Qualitative, order-of-magnitude comparison for biological and organic radical-pair systems. Distances and coupling ranges are approximate; dipolar coupling scales as D/h ~ 52 MHz / r(nm)^3, while exchange is system-specific and often model- or fit-dependent.

System	Radical Pair	Biological / Engineered Use	Pair Distance	Coupling Regime	Hyperfine Landscape	Field Sensitivity
Cryptochrome	FAD radical anion / TrpH radical cation	Biological magnetoreception candidate; flavoprotein signaling model.	~1-2 nm	Moderate dipolar coupling; weak-to-moderate exchange, strongly geometry dependent.	Many 1H and 14N couplings; dominant couplings roughly 0.1-5 mT; appreciable flavin/N-site anisotropy.	Proposed geomagnetic sensitivity; commonly discussed near ~50 microT and tested through mT-scale perturbations.
DNA photolyase	FADH radical / CPD radical anion	Light-driven DNA repair through substrate-bound electron transfer.	~0.5-1 nm	Stronger short-range dipolar coupling; exchange likely larger than cryptochrome; fast ET/recombination window.	Flavin plus damaged-base nuclei; dominant couplings roughly 0.1-10 mT; moderate anisotropy.	Magnetic response is not the central biological function; fast chemistry likely limits weak-field sensitivity.
Photosynthetic reaction center	P+HA- / P+QA- / P700+A1-	Primary photosynthetic charge separation; mature spin-spectroscopy benchmark.	~1-3 nm	Distance-dependent dipolar coupling from tens to few MHz; exchange often small for longer-lived separated pairs.	Pigment and quinone nuclear environments; dominant couplings roughly 0.1-5 mT; moderate anisotropy, well characterized by EPR/CIDNP.	Excellent lab-field spin benchmark; magnetic effects generally studied outside the geomagnetic-compass context.
Carotenoid-porphyrin-fullerene triad	Carotenoid radical cation / fullerene radical anion	Biomimetic organic compass and synthetic radical-pair benchmark.	~2-4 nm	Small dipolar coupling and weak exchange by design in the long-lived charge-separated state.	Asymmetric landscape: carotenoid hyperfine-rich, fullerene hyperfine-poor; dominant couplings ~0-2 mT; low-to-moderate anisotropy.	Clear low-field and mT-scale organic benchmark; geometry and lifetime are tunable by molecular design.
AsLOV2 WT	Triplet FMN / C450 contact intermediate leading to cysteinyl-flavin adduct	Optogenetic protein actuation; flavin-cysteine photochemistry drives mechanical response.	~0.3-1 nm	Very strong contact/near-contact dipolar coupling; strong exchange; short-lived spin intermediate.	FMN plus cysteine/local protein nuclei; dominant couplings roughly 0.1-10 mT; moderate-to-high anisotropy.	Probably not weak-field optimized; useful as a strong-coupling, short-distance extreme.
MagLOV	FMN radical anion / protein donor radical cation; later FMNH radical chemistry	Engineered AsLOV2-derived fluorescent RPM/ODMR sensor for live-cell magnetic readout.	~0.5-2 nm	Engineered spin-correlated radical pair; likely weaker than WT AsLOV2 contact pair, but donor geometry not fully pinned.	Flavin-dominated plus aromatic/protein donor nuclei; dominant couplings roughly 0.1-10 mT; likely moderate-to-high anisotropy.	Large MFE from 0-10 mT; ODMR near ~20 mT with RF around ~600 MHz; strong phenomenological sensor evidence.

Note: Hyperfine ranges summarize dominant couplings, not exhaustive spin Hamiltonians. Field sensitivity indicates literature-relevant regimes and feasibility, not a single performance metric.